Group 39 Project Step 2 Draft

Names

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Project Name

Noir Cards Database Design

Overview

Noir Cards is a locally owned trading card retailer. They sell thousands of different products, most being single cards without a traditional SKU and with fluctuating pricing. Traditional retail products may have less than a dozen on hand at a time, while Noir Cards may carry thousands of a single card at a time. Additionally, they support the local gaming community of approximately 1000 individuals; Tracking customer trends is vital to better recommend products and to encourage brand loyalty. They average between 100-500 transactions per day, and managing their complex inventory is critical for both fulfilling customer orders and preserving client experience. Noir Cards needs an efficient database to record their current inventory, track sales (over $1,000,000 annual revenue), and trace customer information.

Database Description

Products

Records the specifications of products the business carries (single cards and sealed products).

- productID: int, AUTO_INCREMENT, UNIQUE, NOT NULL, PK
- name: varchar(255), NOT NULL
- currentPrice: decimal(12,2), NOT NULL

Relationships:

- **M:N with Inventories** implemented via **ProductInventories** junction table (productID FK).

Inventories

Represents stock locations (display case, stockroom, in transit).

- inventoryID: AUTO_INCREMENT, UNIQUE, NOT NULL, PK
- name: varchar(255), NOT NULL
- atStore: boolean, NOT NULL

Relationships:

- **M:N with Products** implemented via **ProductInventories** junction table (inventoryID FK).

ProductInventories (junction)

Tracks how much of each product exists in each location.

- productInventoryID: int, AUTO_INCREMENT, UNIQUE, NOT NULL, PK
- productID: int, NOT NULL, FK → Products(productID)
- inventoryID: int, NOT NULL, FK → Inventories(inventoryID)
- quantity: int, NOT NULL, default 0

Customers

Stores customer details and loyalty points.Customers are able to sign up for loyalty points and the email list on the website before making a purchase, so they may not have an existing order yet, but are present in the database.

- customerID: int, AUTO_INCREMENT, UNIQUE, NOT NULL, PK
- fName: varchar(255),NOT NULL
- lName: varchar(255), NOT NULL
- email: varchar(255), NOT NULL, UNIQUE
- phone: varchar(15)
- points: int, NOT NULL, DEFAULT 0
- lastPurchase: datetime

Relationships:

- **1:M with Orders**, implemented by customerID FK in Orders.

Orders

One row per transaction.

- orderID: int, AUTO_INCREMENT, UNIQUE, NOT NULL, PK
- customerID: int, FK → Customers(customerID)
- orderDate: datetime, NOT NULL, DEFAULT CURRENT_TIMESTAMP
- orderTotal: decimal(12,2), NOT NULL
- pointsUsed: int, DEFAULT 0

Relationships:

- **1:M with OrderItems** (orderID FK in OrderItems)
- **M:1 with Customers** (customerID FK in Orders)

OrderProducts (junction)

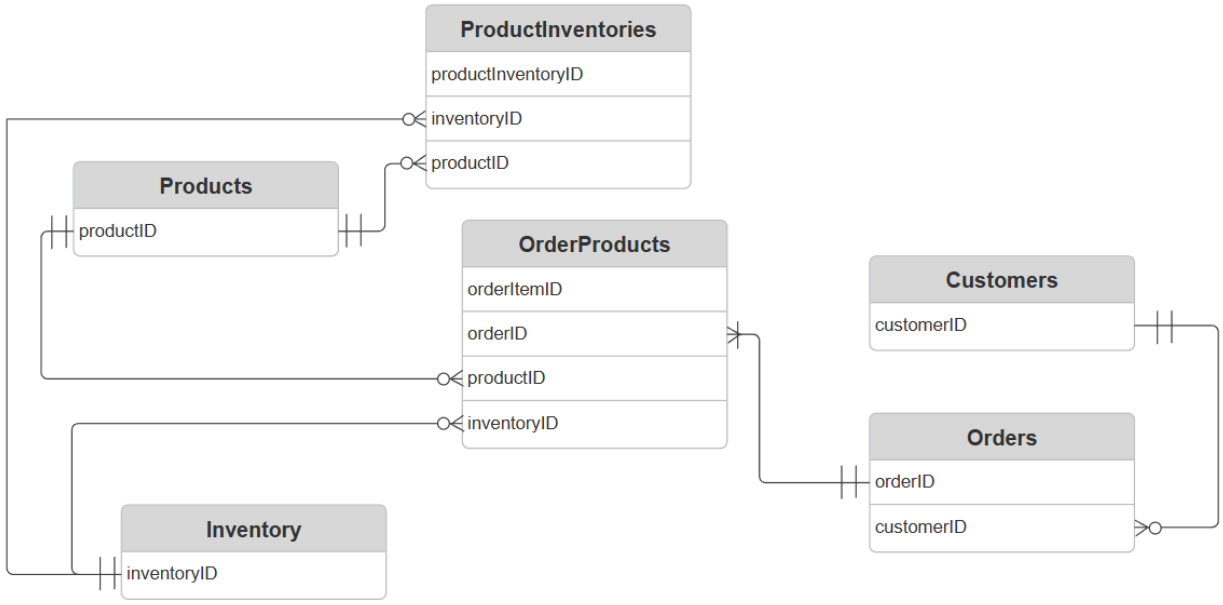
Line items for each order (what products were bought).

- orderItemID: int, AUTO_INCREMENT, UNIQUE, NOT NULL, PK
- orderID: int, NOT NULL → Orders(orderID)
- productID: int, NOT NULL, FK → Products(productID)
- quantity: int, NOT NULL
- inventoryID: int, NOT NULL, FK → Inventories(inventoryID)

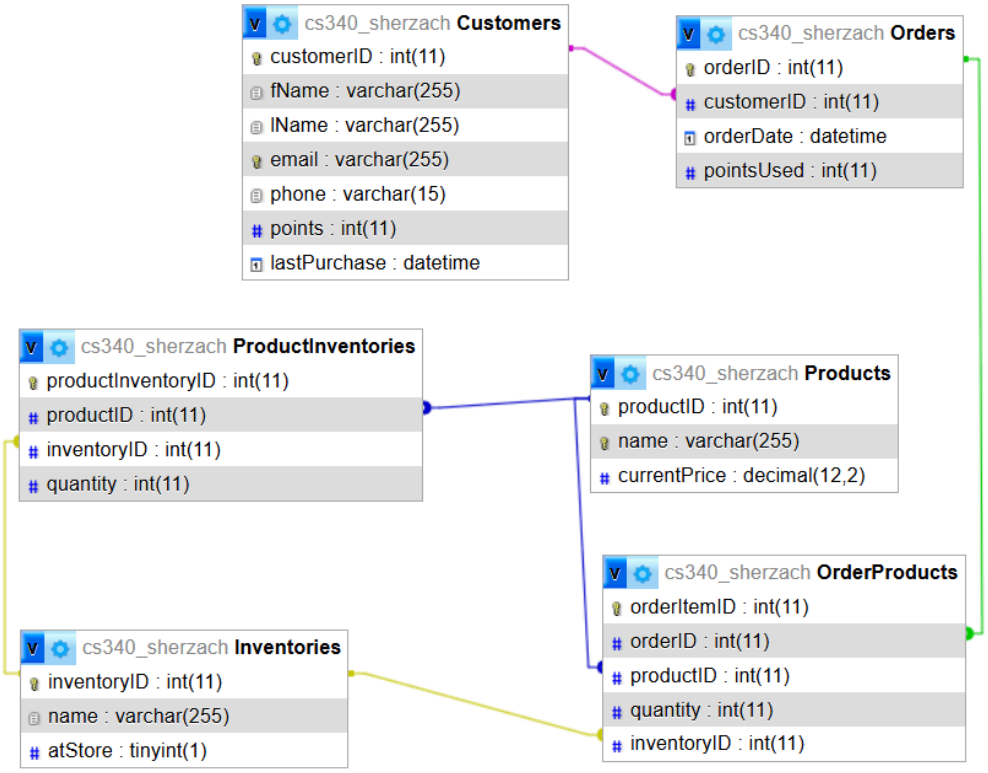
Relationships:

- Orders 1:M OrderItems
- Products 1:M OrderItems
- Inventories 1:M OrderItems

Entity Relationship Diagram:


ProductInventories
productInventoryID
inventoryID
productID
Products
productID
OrderProducts
orderItemID
orderID
productID
inventoryID
Customers
customerID
Orders
orderID
customerID
Inventory
inventoryID

SCHEMA:



Example Data:

For the following example data, any values in parentheses are intended to be helpful identifiers of foreign key entries. They are solely meant to aid comprehension and readability and are not a part of the actual record.

Products:

productID (PK)	name	currentPrice
1	Raging Goblin	2.99
2	Fossil Booster Box	49.99
3	Rhystic Study	1
4	Fog	0.23
5	WOTC branded playmat	17.4

Inventories:

inventoryID (PK)	name	atStore
1	Display	1
2	Store Room	1
3	On Order	0

ProductInventories:

productInventoryID (PK)	productID (FK -> Products.productID)	inventoryID (FK -> Inventory.inventoryID)	quantity
1	1 (Raging Goblin)	1 (Display)	16
2	2 (Fossil Booster Box)	2 (Store Room)	3
3	4 (Fog)	3 (Store Room)	362
4	3 (Rhystic Study)	3 (On Order)	5

Customers:

customerID (PK)	fName	lName	email	phone	points	lastPurchase
1	Henry	Paul	hpaul@email.com	NULL	2000	CURRENTDATE
2	Jason	Isaacs	hizzy@me.net	555-343-5288	200	CURRENTDATE
3	Monica	Isfran	spamfolder@yahoo.com	5554876952	0	CURRENTDATE
4	Mokoto	Nagano	ninjawarrior@super.gove	NULL	3000	CURRENTDATE

Orders:

orderID (PK)	customerID (FK -> Customers.customerID)	orderDate	pointsUsed
1	1 (Henry Paul)	Current Date	1600
2	NULL	Current Date	0
3	2 (Jason Isaacs)	Current Date	200
4	4 (Mokoto Nagano)	Current Date	3000

OrderProducts:

orderItemID (PK)	orderID (FK -> Orders.orderID)	productID (FK -> Products.productID)	quantity	inventoryID (FK -> Inventory.inventoryID)
1	3	4 (Fog)	13	2 (Store Room)
2	1	1 (Raging Goblin)	1	1 (Display)
3	4	2 (Fossil Booster Box)	2	2 (Store Room)

Updates based on feedback from step 2

The primary feedback we received in earlier steps was centered around our ERD. Unfortunately, we did not update our ERD when we made changes in step 2, so it did not reflect all the changes we made. We have since updated our ERD to match our outline and schema exactly. We also received feedback regarding type mismatches between our schema and our outline. We had originally slated the customer data to use VARCHAR(80) for most of the entries, but we made the choice to expand these fields to VARCHAR(255). We have since updated our outline to reflect this.

One reviewer made an excellent point that *Customers.lastPurchase* is technically a derived value from Orders. This technically means that our database has redundancy. However, we have elected to keep this as an attribute of the customer entry. We believe that queries that need this information may have to perform costly searches and filters to verify the date of the most recent transaction. Additionally, this attribute is updated extremely easily when making a new order. Finally, we are not considered with data integrity for this value; Meaning that it is not necessary that this value is definitely true (we don't necessarily care to update this when we also update order details, such as adding a customer to an order).

Citations:

No citations required.